

W1 PRACTICE

From C++ to JS



At the end of this practice, you can

- ✓ Run JS code
- ✓ Create **variables** and **constants**
- ✓ Call and define **functions**
- ✓ Use JS **loops** and **conditions**
- ✓ Manipulate **arrays**, **objects**, **strings**, **Boolean** and **numbers**



Get ready before this practice!

- ✓ **Read** the following documents to understand JS syntax:

<https://cstart.mines.edu/web/Day2/2-JavaScriptBasicSyntax.pdf>

<https://www.integral-domain.org/lwilliams/mis462/JavaScript.pdf>

You can also go further with the following books:

<https://www.gurukultti.org/admin/notice/javascript.pdf>

<https://www.w3schools.com/js/default.asp>

- ✓ **Complete the quiz** (you can re-do it until you have 100% score)



How to submit this practice?

- ✓ **Complete** this document
- ✓ Once finished, join this document to the MS Team assignment and **turn it in**



3 WAYS TO RUN JS CODE

For beginners

To start with, you can just connect to an **online JavaScript editor**, such as this one:

<https://playcode.io/javascript>

For front-end ninjas

Chrome or any other **Web Browser** can execute JavaScript code while loading HTML

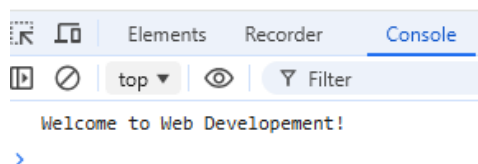
Just create a simple `index.html` file, that links to a `index.js` file:

```
<!DOCTYPE html>
<html>
<head>
  <title>Let's run JS on a Browser</title>
  <script src='index.js'></script>
</head>
<body>
</body>
</html>
```

Then just write some JS code, as example here, we print a message on the Browser console

```
// Example of JS code, printing on console
const courseName = "Web Development";
console.log("Welcome to " + courseName + "!");
```

Finally open your `index.html` on a browser and check the console view



For back-end gurus

Node.js is also able to **execute JavaScript** code **outside a web browser**.

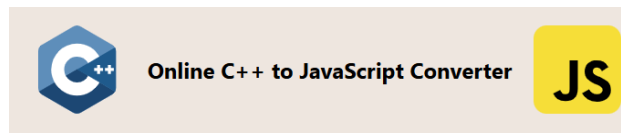
You will need first [to install Node JS](#) on your computer.

You can then just open a terminal on the folder containing your `index.js` file and run

```
node ./index.js
```

PART 1 - UNDERSTAND JS SYNTAX

Note: you can use the [C++ to JS converter](#) to compare C++ and JS syntax.



EXERCISE 1- TYPES, OUTPUTS

Analyze the differences between the provided C++ and JavaScript code.

C++	JS
<pre>#include <iostream> using namespace std; int main() { const int num = 5; for (int i = 0; i < num; i++) { cout << i << " "; } return 0; }</pre>	<pre>const num = 5; for (let i = 0; i < num; i++) { console.log(i); }</pre>

Q1 - What does the **const** key word mean in JS code?

Const is for create a variable that cant be reassigned but the content that is held by the object can be change

Q2 - Why is it necessary **to specify the type** of variables in C++ but not in JavaScript?

C++ requires variable types because it is statically typed and needs type of information at compile time, Js determined the variable types at runtime.

Q3- How to **print in the console** in JS?

node <file-name>.js , if it's about the code its console.log("hello world");

Q4- Is there any difference in the **loop syntax** between C++ and JS?

no

EXERCISE 2 - LOOPS, FUNCTIONS

C++	JS
<pre>#include <iostream> using namespace std; int calculateSum(int array[], int size) { int sum = 0; for (int i = 0; i < size; i++) { // Add here the calculation logic } return sum; } int main() { int arr[] = {1, 2, 3, 4, 5}; cout << calculateSum(arr, 5); return 0; }</pre>	<pre>function calculateSum(array) { let sum = 0; for (let i = 0; i < array.length; i++) { // Add here the calculation logic } return sum; } let arr = [1, 2, 3, 4, 5]; console.log(calculateSum(arr));</pre>

Q1 - Complete the given codes (see comments) to compute the sum of all elements in an array

```
sum += array[i];
```

Q2 – Why the function calculateSum in JS code **does not have the size** parameter?

Because you already put array.length that will give you the length of the array

EXERCISE 3 - CONDITIONS, EQUALITY

JS

```
function myFunction(min, max) {  
  var result = "";  
  for (let number = min; number <= max; number++) {  
    if (number % 2 === 0) {  
      result += number + " - ";  
    }  
  }  
  return result;  
}
```

Q1 – Look at the above code

- Highlight all **variables in blue**
- Underline all **loops in red**
- Highlight all **conditions in green**

Q2 – What is the significance of the modulo operator % in these programs?

It check whether the number is even or not

Q3 – What is the difference between === and == in JS? *Highlight the right answer*

4 == 9	TRUE / FALSE
4 == 4	TRUE / FALSE
4 == "4"	TRUE / FALSE
4 === "4"	TRUE / FALSE

Q4 – What will this code will print on console?

```
console.log(myFunction(9, 14))
```

10 - 12 - 14 -

Q5 – What will this code will print on console?

```
console.log(myFunction(7, 3))
```

Nothing because the var number is bigger than the max and met the condition of the loop

EXERCISE 4 – MEMORY ALLOCATION

Both codes are performing the same job:

C++

```
#include <iostream>
using namespace std;

int main() {
    int size = 5;
    int* arr = new int[size];
    for (int i = 0; i < size; i++) {
        arr[i] = i * 2;
    }

    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    delete[] arr;
    return 0;
}
```

JS

```
let size = 5;
let arr = [];
for (let i = 0; i < size; i++) {
    arr[i] = i * 2;
}

for (let i = 0; i < size; i++) {
    console.log(arr[i]);
}
```

Q1 – In both codes, are we using a **static** or a **dynamic** array? Explain why...

We are using dynamic array in both code. In js the size of the array change grow automatically. In C++ code you reallocated the size of the array with the size var and can always realloc the size of the array later on in the heap.

Q2 – Explain why JavaScript **does not** need explicit **memory allocation** or **deallocation**, as C++ need it

Because Js have build in garbage collection

PART 2 - CODE JS CHALLENGES



Good job!

Now you should know the [basic syntax of JavaScript!](#)

Let's solve some problem now.

Each challenge is structured the same way:

- **Goal** What the function shall do
- **Inputs:** the function parameters
- **Output** the function return

As example, for the challenge 1, you will provide the following function:

```
function challenge1(width, height) {  
  let rectangleString = '';  
  // Your code  
  for(let i = 0; i < height; i++) {  
    for(let j=0; j < width; j++) {  
      rectangleString += '* ';  
    }  
    rectangleString += '\n';  
  }  
  
  return rectangleString;  
}
```

CHALLENGE 1		EASY
Draw a rectangle in the console using stars		
INPUT	OUTPUT	
width 3 height 4	<pre>*** *** *** ***</pre>	
width 5 height 2	<pre>***** *****</pre>	
width 5 height -2		

CHALLENGE 2		MEDIUM
Reverse an array		
INPUT	OUTPUT	
array [14,15,16,20]	[20,16,15,14]	
array [5,4,3,2,1]	[1,2,3,4,5]	
array []	[]	

Any help on arrays with JavaScript? [Check here](#).

```
function challenge2(arr) {
  arr.reverse();
  return arr;
}

console.log(challenge2([1,2,3,4]));
```

CHALLENGE 3		MEDIUM
Calculate the average grade of a list of students.		
INPUT	OUTPUT	
array [85, 90, 78, 92]	86.25	
array [10,20,30]	20	
array []	0	


```
function challenge3(arr) {
  const subjects = arr.length;
  let sum = 0;

  for(let i=0; i < subjects; i++) {
    sum += arr[i];
  }

  return sum / subjects;
}

console.log(challenge3([10,20,30]));
```

CHALLENGE 4		MEDIUM
Write a function to count how many times a character appears in a string.		
INPUT	OUTPUT	
text "hello world" char = 'o'	2	
text "aaa bbb a" char = 'a'	4	
text "abc" char = 'd'	0	

```
function challenge4(string, char) {
  let res = 0;
  for(let i = 0; i < string.length; i++) {
    if(string[i] == char) res++;
  }

  return res;
}

console.log(challenge4("aaa s aaa", "s"));
```

CHALLENGE 5		HARD
Count the number of words in a sentence		
INPUT	OUTPUT	

text "hello world"	2
text "this is the best day"	5
text "a bb ccc ddddddd e"	5

```
function challenge5(string) {
  let res = 0
  for(let i =0; i < string.length; i++) {
    while(string[i] == " ") i++;
    res++ ;
    while(i < string.length && string[i] !== " ") i++;
  }

  return res;
}

console.log(challenge5("tv na   sdsfsdf       ot te"));
```

CHALLENGE 6		HARD
Simulate a voting system for three candidates (A / B/ C). Count votes and declare a winner		
INPUT	OUTPUT	
votes ['A', 'B', 'A', 'C', 'A']	A is the winner	
votes ['A', 'B', 'B', 'C', 'A']	A and B are both winners	
votes []	There is not vote yet	

```
function challenge6(votes) {
  let voteCount = [];
  for(const vote of votes) {
    voteCount[vote] = (voteCount[vote] || 0) +1;
  }

  let max = 0;
  for(const candidate in voteCount) {
    if(voteCount[candidate] > max) max=voteCount[candidate];
  }

  let winner = [];
  for(const candidate in voteCount) {
    if(voteCount[candidate] == max) {
```

```
        winner.push(candidate);
    }
}

return winner;
}

console.log(challenge6(['a','a','b','b']))
```